

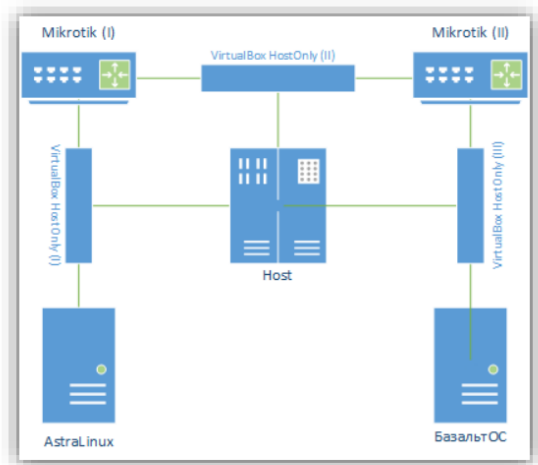
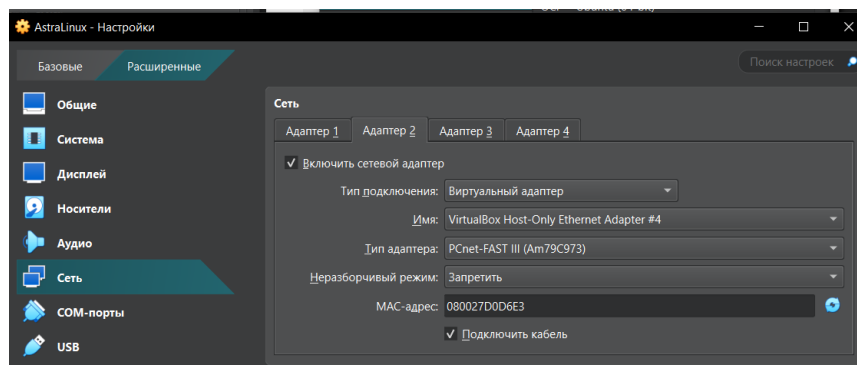
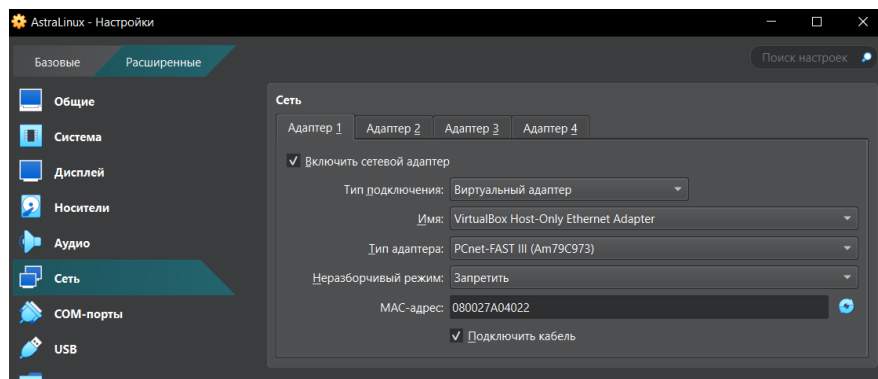


Рисунок 1 – Конфигурация сети для практического занятия

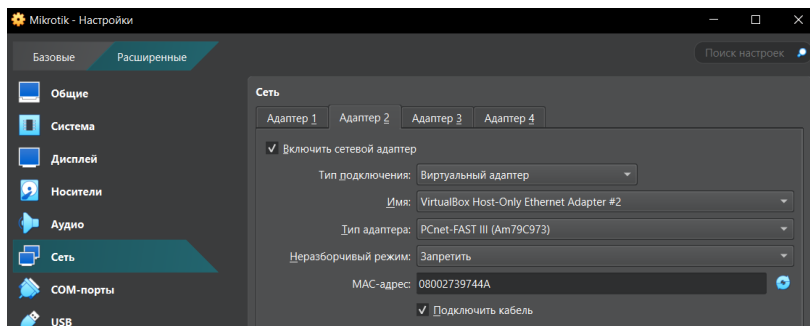
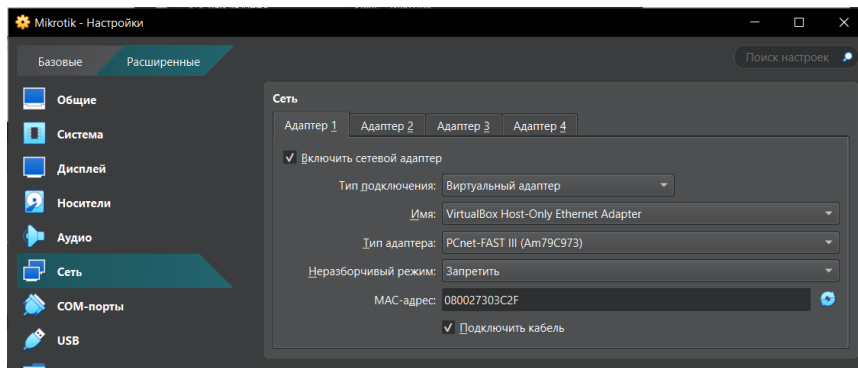
1.

Виртуальные сети			
Имя	IPv4 префикс	IPv6 префикс	DHCP сервер
VirtualBox Host-Only Ethernet Adapter	10.16.65.2/18		Выключен
VirtualBox Host-Only Ethernet Adapter #2	192.168.56.1/24		Выключен
VirtualBox Host-Only Ethernet Adapter #3	192.168.205.1/24		Выключен
VirtualBox Host-Only Ethernet Adapter #4	192.168.201.1/24		Выключен

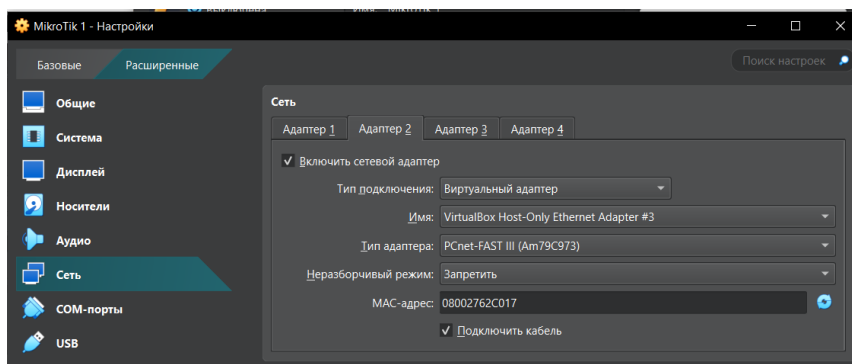
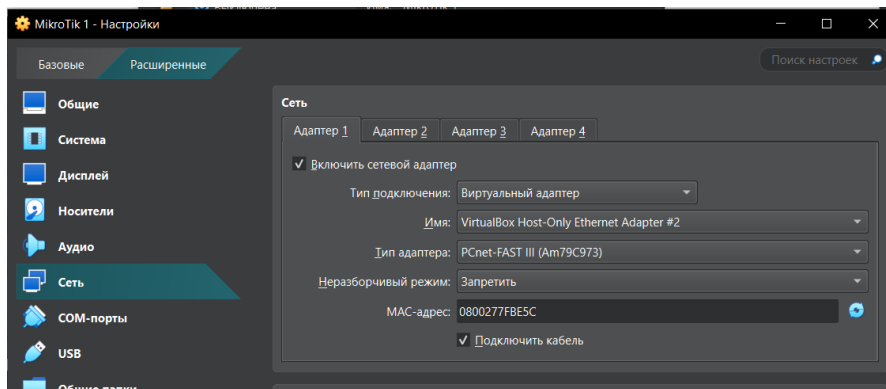
Astra Linux:



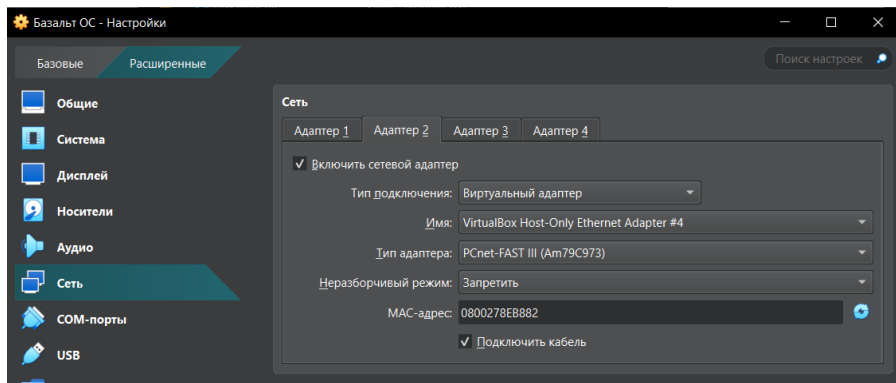
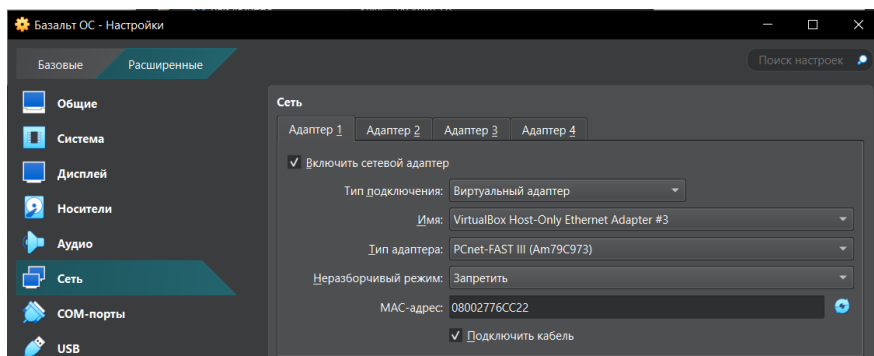
MikroTik:



MikroTik 1:



Базальт-ОС:



2.

Маска: 255.255.255.192 (/26)

Подсеть 1: 10.10.16.0/26

- Диапазон адресов: 10.10.16.0 – 10.10.16.63
- Адрес сети: 10.10.16.0
- Широковещательный адрес: 10.10.16.63
- Доступные адреса: 10.10.16.1 – 10.10.16.62

Подсеть 2: 10.10.16.64/26

- Диапазон адресов: 10.10.16.64 – 10.10.16.127
- Адрес сети: 10.10.16.64
- Широковещательный адрес: 10.10.16.127
- Доступные адреса: 10.10.16.65 – 10.10.16.126

Подсеть 3: 10.10.16.128/26

- Диапазон адресов: 10.10.16.128 – 10.10.16.191
- Адрес сети: 10.10.16.128
- Широковещательный адрес: 10.10.16.191
- Доступные адреса: 10.10.16.129 – 10.10.16.190

Подсеть 4: 10.10.16.192/26

- Диапазон адресов: 10.10.16.192 – 10.10.16.255
- Адрес сети: 10.10.16.192
- Широковещательный адрес: 10.10.16.255
- Доступные адреса: 10.10.16.193 – 10.10.16.254

VirtualBox Host Only 1: 10.10.16.1/26

Astra(eth0):

10.10.16.5/26

Микротик (ether1):

10.10.16.10/26

VirtualBox Host Only 2: 10.10.16.65/26

Микротик (ether2):

10.10.16.66/26

Микротик 1 (ether1):

10.10.16.70/26

VirtualBox Host Only 3: 10.10.16.129/26

Микротик 1 (ether2):

10.10.16.130/26

Базальт (enp0s3):

10.10.16.135/26

VirtualBox Host Only 4: 10.10.16.193/26

Astra (eth1):

10.10.16.195/26

Базальт (enp0s8):

10.10.16.205/26

Виртуальные сети			
Сети NAT			
Облачные сети			
Имя	IPv4 префикс	IPv6 префикс	DHCP сервер
VirtualBox Host-Only Ethernet Adapter	10.10.16.1/26		Выключен
VirtualBox Host-Only Ethernet Adapter #2	10.10.16.65/26		Выключен
VirtualBox Host-Only Ethernet Adapter #3	10.10.16.129/26		Выключен
VirtualBox Host-Only Ethernet Adapter #4	10.10.16.193/26		Выключен

Микротик:

```
Press F1 for help
Podkorytova Aleksandra IP-311

[admin@MikroTik] > ip address/add address=10.10.16.10/26 interface=ether1
[admin@MikroTik] > ip address/add address=10.10.16.66/26 interface=ether2

2 10.10.16.10/26      10.10.16.0      ether1
3 10.10.16.66/26     10.10.16.64     ether2
```

Микротик 1:

```
[admin@MikroTik1] > ip address/add address=10.10.16.70/26 interface=ether2
[admin@MikroTik1] > ip address/add address=10.10.16.130/26 interface=ether3

0 10.10.16.70/26     10.10.16.64     ether2
1 10.10.16.130/26    10.10.16.128    ether3
```

Астра:

```
user@host-astra:~$ sudo ip addr add 10.10.16.5/26 dev eth0
user@host-astra:~$ sudo ip addr add 10.10.16.195/26 dev eth1
user@host-astra:~$
```

```

2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UNKNOWN group def
n 1000
    link/ether 08:00:27:a0:40:22 brd ff:ff:ff:ff:ff:ff
    inet 10.10.16.5/26 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fea0:4022/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UNKNOWN group def
n 1000
    link/ether 08:00:27:d0:d6:e3 brd ff:ff:ff:ff:ff:ff
    inet 10.16.0.8/18 brd 10.16.63.255 scope global eth1
        valid_lft forever preferred_lft forever
    inet 10.10.16.195/26 scope global eth1
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fed0:d6e3/64 scope link
        valid_lft forever preferred_lft forever

```

Базальт:

```

[root@host-basalt ~]# sudo ip addr add 10.10.16.135/26 dev enp0s3
[root@host-basalt ~]# sudo ip addr add 10.10.16.205/26 dev enp0s8
[root@host-basalt ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default q
n 1000
    link/ether 08:00:27:76:cc:22 brd ff:ff:ff:ff:ff:ff
    inet 10.10.16.135/26 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe76:cc22/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 08:00:27:8e:b8:82 brd ff:ff:ff:ff:ff:ff
    inet 10.10.16.205/26 scope global enp0s8
        valid_lft forever preferred_lft forever

```

3.

Астра:

Добавление маршрута для подсети 10.10.16.64/26 (VirtualBox Host Only 2):

```

user@host-astra:~$ sudo ip route add 10.10.16.64/26 via 10.10.16.10
user@host-astra:~$ ip route
default via 10.16.0.1 dev eth1 onlink
10.10.16.0/26 dev eth0 proto kernel scope link src 10.10.16.5
10.10.16.64/26 via 10.10.16.10 dev eth0
10.10.16.192/26 dev eth1 proto kernel scope link src 10.10.16.195
10.16.0.0/18 dev eth1 proto kernel scope link src 10.16.0.8
user@host-astra:~$

```

- **10.10.16.64/26:** Целевая подсеть (VirtualBox Host Only 2)
- **via 10.10.16.10:** Адрес следующего (MikroTik (ether1): 10.10.16.10)

Микротик:

Добавление маршрута для подсети 10.10.16.128/26 (VirtualBox Host Only 3):

```
Podkorytova Aleksandra IP-311
[admin@MikroTik] > ip route add dst-address=10.10.16.128/26 gateway=10.10.16.70
[admin@MikroTik] > _
```

- **dst-address:** Целевая подсеть (10.10.16.128/26)
- **gateway:** Адрес следующего (MikroTik 1, ether2: 10.10.16.70)

Добавление маршрута для подсети 10.10.16.192/26 (VirtualBox Host Only 4):

```
[admin@MikroTik] > ip route add dst-address=10.10.16.192/26 gateway=10.10.16.70
[admin@MikroTik] > _
```

- **dst-address:** Целевая подсеть (10.10.16.192/26)
- **gateway:** Адрес следующего (MikroTik 1, ether2: 10.10.16.70)

```
[admin@MikroTik] > ip route print
Flags: D - DYNAMIC; A - ACTIVE; c - CONNECT, s - STATIC
Columns: DST-ADDRESS, GATEWAY, DISTANCE
#    DST-ADDRESS    GATEWAY    DISTANCE
0    DAc 10.10.16.0/26 ether1      0
0    DAc 10.10.16.64/26 ether2      0
0    As 10.10.16.128/26 10.10.16.70 1
1    As 10.10.16.192/26 10.10.16.70 1
    DAc 10.16.64.0/18 ether1      0
    DAc 192.168.56.0/24 ether2      0
[admin@MikroTik] >
```

Микротик 1:

Добавление маршрута для подсети 10.10.16.0/26 (VirtualBox Host Only 1):

```
[admin@MikroTik1] > ip route add dst-address=10.10.16.0/26 gateway=10.10.16.66
[admin@MikroTik1] > _
```

- **dst-address:** Целевая подсеть (10.10.16.0/26)
- **gateway:** Адрес следующего (MikroTik, ether2: 10.10.16.66)

Добавление маршрута для подсети 10.10.16.192/26 (VirtualBox Host Only 4):

```

[admin@MikroTik1] > ip route add dst-address=10.10.16.192/26 gateway=10.10.16.205
[admin@MikroTik1] > ip route print
Flags: D - DYNAMIC; I - INACTIVE, A - ACTIVE; c - CONNECT, s - STATIC; H - HW-OF
FLOADED
Columns: DST-ADDRESS, GATEWAY, DISTANCE
# DST-ADDRESS GATEWAY DISTANCE
0 As 10.10.16.0/26 10.10.16.66 1
DAc 10.10.16.64/26 ether2 0
DAc 10.10.16.128/26 ether3 0
1 IsH 10.10.16.192/26 10.10.16.205 1

```

- dst-address: Целевая подсеть (10.10.16.192/26)
- gateway: Адрес следующего (БазальтОС, enp0s8: 10.10.16.205)

Базальт:

Добавление маршрута для подсети 10.10.16.0/26 (VirtualBox Host Only 1):

```
[root@host-basalt ~]# ip route add 10.10.16.0/26 via 10.10.16.130
```

о 10.10.16.0/26: Целевая подсеть (VirtualBox Host Only 1).

о via 10.10.16.130: Адрес следующего хопа (MikroTik 1, ether3: 10.10.16.130)

Добавление маршрута для подсети 10.10.16.64/26 (VirtualBox Host Only 2):

```
[root@host-basalt ~]# ip route add 10.10.16.64/26 via 10.10.16.130
```

о 10.10.16.64/26: Целевая подсеть (VirtualBox Host Only 2)

о via 10.10.16.130: Адрес следующего хопа (MikroTik 1, ether3: 10.10.16.130)

```

[root@host-basalt ~]# ip route
10.10.16.0/26 via 10.10.16.130 dev enp0s3
10.10.16.64/26 via 10.10.16.130 dev enp0s3
10.10.16.128/26 dev enp0s3 proto kernel scope link src 10.10.16.135

```

Проверка связанности между узлами:

Астра ⇔ Микротик


```
Press F1 for help
Podkorytova Aleksandra IP-311

[admin@MikroTik] > ping 10.10.16.5
  SEQ HOST                                SIZE TTL TIME          STATUS
  0 10.10.16.5                            56 64 1ms229us
  1 10.10.16.5                            56 64 1ms101us
  2 10.10.16.5                            56 64 1ms361us
  3 10.10.16.5                            56 64 1ms522us
  4 10.10.16.5                            56 64 1ms363us
  5 10.10.16.5                            56 64 1ms221us
  6 10.10.16.5                            56 64 1ms173us
sent=7 received=7 packet-loss=0% min-rtt=1ms101us avg-rtt=1ms281us
max-rtt=1ms522us

[admin@MikroTik] >

user@host-astra:~$ ping 10.10.16.10
PING 10.10.16.10 (10.10.16.10) 56(84) bytes of data.
 64 bytes from 10.10.16.10: icmp_seq=1 ttl=64 time=2.22 ms
 64 bytes from 10.10.16.10: icmp_seq=2 ttl=64 time=0.872 ms
 64 bytes from 10.10.16.10: icmp_seq=3 ttl=64 time=0.620 ms
 64 bytes from 10.10.16.10: icmp_seq=4 ttl=64 time=1.08 ms
^C
--- 10.10.16.10 ping statistics ---
 4 packets transmitted, 4 received, 0% packet loss, time 3005ms
 rtt min/avg/max/mdev = 0.620/1.199/2.222/0.613 ms
user@host-astra:~$ ping 10.10.16.66
PING 10.10.16.66 (10.10.16.66) 56(84) bytes of data.
 64 bytes from 10.10.16.66: icmp_seq=1 ttl=64 time=1.50 ms
 64 bytes from 10.10.16.66: icmp_seq=2 ttl=64 time=1.05 ms
 64 bytes from 10.10.16.66: icmp_seq=3 ttl=64 time=1.70 ms
^Z
[1]+  Остановлен ping 10.10.16.66
user@host-astra:~$
```

Микротик ⇄ Микротик1

```
MikroTik [Работает] - Oracle VirtualBox
Файл  Машина  Вид  Ввод  Устройства  Справка
  4 10.10.16.5                            56 64 1ms363us
  5 10.10.16.5                            56 64 1ms221us
  6 10.10.16.5                            56 64 1ms173us
sent=7 received=7 packet-loss=0% min-rtt=1ms101us avg-rtt=1ms281us
max-rtt=1ms522us

[admin@MikroTik] > ping 10.10.16.70
  SEQ HOST                                SIZE TTL TIME          STATUS
  0 10.10.16.70                            56 64 2ms12us
  1 10.10.16.70                            56 64 1ms90us
  2 10.10.16.70                            56 64 766us
sent=3 received=3 packet-loss=0% min-rtt=766us avg-rtt=1ms289us
max-rtt=2ms12us

[admin@MikroTik] > ping 10.10.16.130\
^C
[admin@MikroTik] > ping 10.10.16.130
  SEQ HOST                                SIZE TTL TIME          STATUS
  0 10.10.16.130                           56 64 1ms119us
  1 10.10.16.130                           56 64 1ms601us
  2 10.10.16.130                           56 64 1ms377us
sent=3 received=3 packet-loss=0% min-rtt=1ms119us avg-rtt=1ms365us
max-rtt=1ms601us

[admin@MikroTik] >
```

```
MikroTik 1 [Работает] - Oracle VirtualBox
Файл  Машина  Вид  Ввод  Устройства  Справка
МММ  МММ  МММ  ККК  ККК  РРР  РРР  000000  ТТТ  ТТТ  ККК  ККК
MikroTik RouterOS 7.16.2 (c) 1999-2024 https://www.mikrotik.com/

Press F1 for help

[admin@MikroTik1] > ping 10.10.16.10
  SEQ HOST                                SIZE TTL TIME          STATUS
  0 10.10.16.10                            56 64 1ms444us
  1 10.10.16.10                            56 64 1ms24us
  2 10.10.16.10                            56 64 977us
sent=3 received=3 packet-loss=0% min-rtt=977us avg-rtt=1ms148us
max-rtt=1ms444us

[admin@MikroTik1] > ping 10.10.16.66
  SEQ HOST                                SIZE TTL TIME          STATUS
  0 10.10.16.66                            56 64 1ms422us
  1 10.10.16.66                            56 64 1ms24us
  2 10.10.16.66                            56 64 1ms85us
  3 10.10.16.66                            56 64 1ms79us
sent=4 received=4 packet-loss=0% min-rtt=1ms24us avg-rtt=1ms152us
max-rtt=1ms422us

[admin@MikroTik1] >
```

Микротик1 ⇄ Базальт

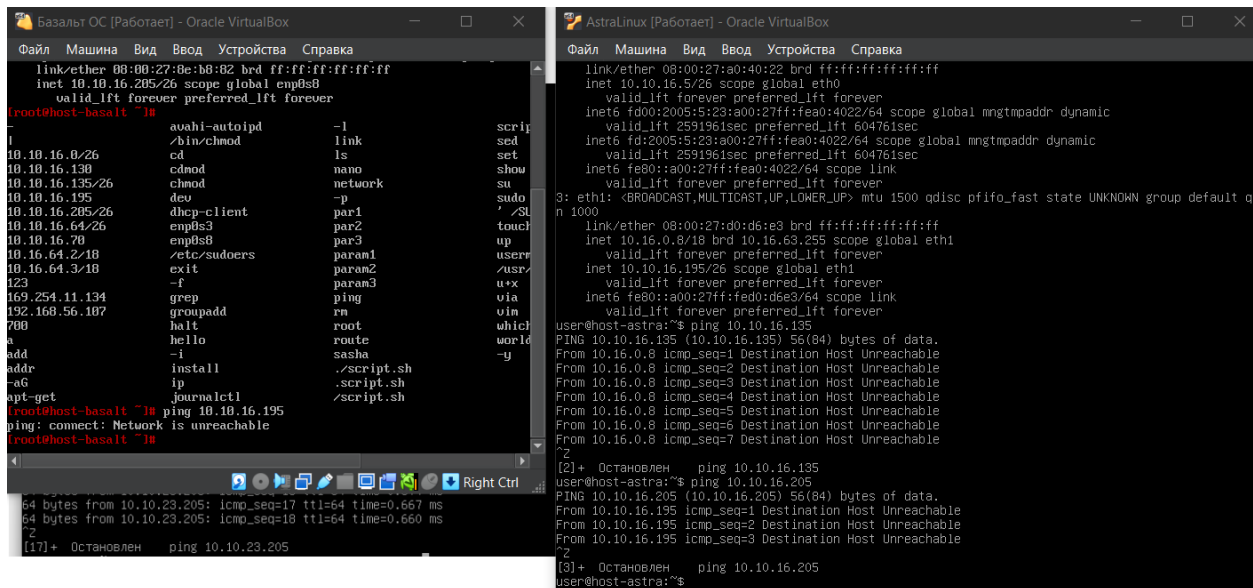
```
[root@host-basalt ~]# ping 10.10.16.195
ping: connect: Network is unreachable
[root@host-basalt ~]# ping 10.10.16.130
PING 10.10.16.130 (10.10.16.130) 56(84) bytes of data.
 64 bytes from 10.10.16.130: icmp_seq=1 ttl=64 time=1.42 ms
 64 bytes from 10.10.16.130: icmp_seq=2 ttl=64 time=0.992 ms
^C
--- 10.10.16.130 ping statistics ---
 2 packets transmitted, 2 received, 0% packet loss, time 1002ms
 rtt min/avg/max/mdev = 0.992/1.206/1.420/0.214 ms
[root@host-basalt ~]# ping 10.10.16.70
PING 10.10.16.70 (10.10.16.70) 56(84) bytes of data.
 64 bytes from 10.10.16.70: icmp_seq=1 ttl=64 time=0.946 ms
 64 bytes from 10.10.16.70: icmp_seq=2 ttl=64 time=1.09 ms
 64 bytes from 10.10.16.70: icmp_seq=3 ttl=64 time=1.03 ms
^C
--- 10.10.16.70 ping statistics ---
 3 packets transmitted, 3 received, 0% packet loss, time 2004ms
 rtt min/avg/max/mdev = 0.946/1.019/1.086/0.057 ms
[root@host-basalt ~]#
```

```
Columns: DST-ADDRESS, GATEWAY, DISTANCE
# DST-ADDRESS GATEWAY DISTANCE
0 As 10.10.16.0/26 10.10.16.66 1
Dac 10.10.16.64/26 ether2 0
Dac 10.10.16.128/26 ether3 0
1 IsH 10.10.16.192/26 10.10.16.205 1
[admin@MikroTik1] > ping 10.10.16.205
  SEQ HOST                                SIZE TTL TIME          STATUS
  0
  1
  2
sent=3 received=0 packet-loss=100%

[admin@MikroTik1] > ping 10.10.16.135
  SEQ HOST                                SIZE TTL TIME          STATUS
  0 10.10.16.135                           56 64 985us
  1 10.10.16.135                           56 64 997us
sent=2 received=2 packet-loss=0% min-rtt=985us avg-rtt=991us
max-rtt=997us

[admin@MikroTik1] >
```

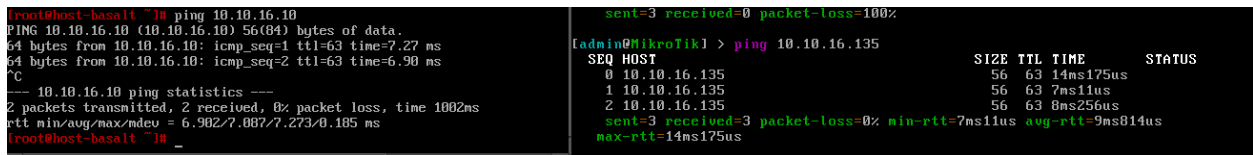
Базальт ⇄ Астра (почему-то нет связи) НАСТРОИТЬ СВЯЗЬ



Астра ↔ Микротик1



Базальт ↔ Микротик



4.

Исходный выделенный префикс:

fd00:2005:05::/48

Разбиение на подсети:

1. fd00:2005:05:1000::/50 (Astralinux → vboxnet1 → mikrotik-01)
2. fd00:2005:05:4000::/50 (mikrotik-01 → vboxnet2 → mikrotik-02)
3. fd00:2005:05:8000::/50 (mikrotik-02 → vboxnet3 → БазальтОС)
4. fd00:2005:05:C000::/50 (БазальтОС → vboxnet4 → Astralinux)

Настройка интерфейсов и адресов:

Микротик:

ipv6 address add address=fd00:2005:05:1000::1/50 interface=ether1

ipv6 address add address=fd00:2005:05:4000::1/50 interface=ether2\

```
Podkorytova Aleksandra IP-311
[admin@MikroTik] > ipv6 address add address=fd:2005:05:1000::1/50 interface=ether1
[admin@MikroTik] > ipv6 address add address=fd:2005:05:4000::1/50 interface=ether2
[admin@MikroTik] > ipv6 address print
Flags: D - DYNAMIC; G - GLOBAL, L - LINK-LOCAL
Columns: ADDRESS, INTERFACE, ADVERTISE
# ADDRESS INTERFACE ADVERTISE
0 G fd00:2005:5:23::1/64 ether1 yes
1 G fd:2005:5:23::1/64 ether1 yes
2 D ::1/128 lo no
3 DL fe80::a00:27ff:fe30:3c2f/64 ether1 no
4 DL fe80::a00:27ff:fe39:744a/64 ether2 no
5 G fd:2005:5:1000::1/50 ether1 no
6 G fd:2005:5:4000::1/50 ether2 no
[admin@MikroTik] >
```

Микротик 1:

ipv6 address add address=fd00:2005:05:4000::2/50 interface=ether2

ipv6 address add address=fd00:2005:05:8000::1/50 interface=ether3

```
Press F1 for help
[admin@MikroTik1] > ipv6 address add address=fd:2005:05:4000::2/50 interface=ether2
[admin@MikroTik1] > ipv6 address add address=fd:2005:05:8000::2/50 interface=ether3
[admin@MikroTik1] > ipv6 address print
Flags: D - DYNAMIC; G - GLOBAL, L - LINK-LOCAL
Columns: ADDRESS, INTERFACE, ADVERTISE
# ADDRESS INTERFACE ADVERTISE
0 D ::1/128 lo no
1 DL fe80::a00:27ff:fe7f:be5c/64 ether2 no
2 DL fe80::a00:27ff:fe62:c017/64 ether3 no
3 G fd:2005:5:4000::2/50 ether2 no
4 G fd:2005:5:8000::2/50 ether3 no
[admin@MikroTik1] >
```

Базальт-ОС:

sudo ip -6 addr add fd00:2005:05:8000::3/50 dev enp0s3

sudo ip -6 addr add fd00:2005:05:c000::1/50 dev enp0s8

```
[root@host-basalt ~]# sudo ip -6 addr add fd00:2005:05:8000::3/50 dev enp0s3
[root@host-basalt ~]# sudo ip -6 addr add fd00:2005:05:c000::1/50 dev enp0s8
[root@host-basalt ~]# ip -6 a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 state UNKNOWN qlen 1000
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UNKNOWN qlen 1000
    inet6 fd00:2005:5:8000::3/50 scope global
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe76:cc22/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST> mtu 1500 state DOWN qlen 1000
    inet6 fd00:2005:5:c000::1/50 scope global tentative
        valid_lft forever preferred_lft forever
[root@host-basalt ~]#
```

Включила интерфейс enp0s8:

```
[root@host-basalt ~]# sudo ip link set enp0s8 up
[root@host-basalt ~]# ip -6 up
Object "up" is unknown, try "ip help".
[root@host-basalt ~]# ip -6 a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 state UNKNOWN qlen 1000
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UNKNOWN qlen 1000
    inet6 fd00:2005:5:8000::3/50 scope global
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe76:cc22/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UNKNOWN qlen 1000
    inet6 fd00:2005:5:c000::1/50 scope global
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe0e:b882/64 scope link
        valid_lft forever preferred_lft forever
```

Астра Линукс:

sudo ip -6 addr add fd00:2005:05:c000::2/50 dev eth1

sudo ip -6 addr add fd00:2005:05:1000::2/50 dev eth0

(префиксы стоят на первом месте у каждого из адаптеров)

```
user@host-astra:~$ sudo ip -6 addr add fd00:2005:05:c000::2/50 dev eth1
user@host-astra:~$ sudo ip -6 addr add fd00:2005:05:1000::2/50 dev eth0
user@host-astra:~$ ip -6 a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 state UNKNOWN qlen 1000
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UNKNOWN qlen 1000
    inet6 fd00:2005:5:1000::2/50 scope global
        valid_lft forever preferred_lft forever
    inet6 fd00:2005:5:23:a00:27ff:fea0:4022/64 scope global mngtmpaddr dynamic
        valid_lft 2591813sec preferred_lft 604613sec
    inet6 fd:2005:5:23:a00:27ff:fea0:4022/64 scope global mngtmpaddr dynamic
        valid_lft 2591813sec preferred_lft 604613sec
    inet6 fe80::a00:27ff:fea0:4022/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UNKNOWN qlen 1000
    inet6 fd00:2005:5:c000::2/50 scope global
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fed0:d6e3/64 scope link
        valid_lft forever preferred_lft forever
user@host-astra:~$
```

Настройка маршрутов:

Микротик:

/ipv6 route

add dst-address=fd00:2005:05:8000::/50 gateway=fd00:2005:05:4000::2

add dst-address=fd00:2005:05:c000::/50 gateway=fd00:2005:05:4000::2

```
[admin@MikroTik] > ipv6/route/add dst-address=fd00:2005:05:8000::/50 gateway=fd00:2005:05:4000::2
[admin@MikroTik] > ipv6/route/add dst-address=fd00:2005:05:c000::/50 gateway=fd00:2005:05:4000::2
[admin@MikroTik] > ipv6/route/print
Flags: D - DYNAMIC; I - INACTIVE; A - ACTIVE; c - CONNECT, s - STATIC; H - HW-OF
FLOADED
Columns: DST-ADDRESS, GATEWAY, DISTANCE
# DST-ADDRESS GATEWAY DISTANCE
0 DAc ::1/128 lo 0
0 DAc fd:2005:5::/50 ether1 0
0 DAc fd:2005:5:23::/64 ether1 0
0 DAc fd:2005:5:4000::/50 ether2 0
0 DAc fd00:2005:5:23::/64 ether1 0
0 IsH fd00:2005:5:8000::/50 fd00:2005:5:4000::2 1
1 IsH fd00:2005:5:c000::/50 fd00:2005:5:4000::2 1
0 DAc fe80::zether1/64 ether1 0
0 DAc fe80::zether2/64 ether2 0
[admin@MikroTik] >
```

Микротик 1:

/ipv6 route

add dst-address=fd00:2005:05:1000::/50 gateway=fd00:2005:05:4000::1

add dst-address=fd00:2005:05:c000::/50 gateway=fd00:2005:05:8000::2

```
[admin@MikroTik] > ipv6/route/add dst-address=fd00:2005:05:1000::/50 gateway=fd00:2005:05:4000::1
[admin@MikroTik] > ipv6/route/add dst-address=fd00:2005:05:c000::/50 gateway=fd00:2005:05:8000::2
[admin@MikroTik] > ipv6/route/print
Flags: D - DYNAMIC; I - INACTIVE; A - ACTIVE; c - CONNECT, s - STATIC; H - HW-OF
FLOADED
Columns: DST-ADDRESS, GATEWAY, DISTANCE
# DST-ADDRESS GATEWAY DISTANCE
0 DAc ::1/128 lo 0
0 DAc fd:2005:5:4000::/50 ether2 0
0 DAc fd:2005:5:8000::/50 ether3 0
0 IsH fd00:2005:5::/50 fd00:2005:5:4000::1 1
1 IsH fd00:2005:5:c000::/50 fd00:2005:5:8000::2 1
0 DAc fe80::zether2/64 ether2 0
0 DAc fe80::zether3/64 ether3 0
[admin@MikroTik] >
```

Базальт-ОС:

ip -6 route add fd00:2005:05:1000::/50 via fd00:2005:05:8000::1

```
[root@host-basalt ~]# ip -6 route add fd00:2005:05:1000::/50 via fd00:2005:05:8000::1
[root@host-basalt ~]# ip -6 a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 state UNKNOWN qlen 1000
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UNKNOWN qlen 1000
    inet6 fd00:2005:5:8000::3/50 scope global
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe76:cc22/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UNKNOWN qlen 1000
    inet6 fd00:2005:5:c000::1/50 scope global
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe8e:b882/64 scope link
        valid_lft forever preferred_lft forever
[root@host-basalt ~]# ip -6 r
fd00:2005:5::/50 via fd00:2005:5:8000::1 dev enp0s3 metric 1024 pref medium
fd00:2005:5:8000::/50 dev enp0s3 proto kernel metric 256 pref medium
fd00:2005:5:c000::/50 dev enp0s8 proto kernel metric 256 pref medium
fe80::/64 dev enp0s3 proto kernel metric 256 pref medium
fe80::/64 dev enp0s8 proto kernel metric 256 pref medium
[root@host-basalt ~]#
```

Астра Линукс:

sudo ip -6 route add fd00:2005:05:1000::/50 via fd00:2005:05:1000::1

```
user@host-astra:~$ sudo ip -6 route add fd00:2005:05:1000::/50 via fd00:2005:05:1000::1
user@host-astra:~$ ip -6 r
fd:2005:5:23::/64 dev eth0 proto kernel metric 256 expires 2591640sec pref medium
fd00:2005:5:23::/64 dev eth0 proto kernel metric 256 expires 2591640sec pref medium
fd00:2005:5::/50 dev eth0 proto kernel metric 256 pref medium
fd00:2005:5::/50 via fd00:2005:5:1000::1 dev eth0 metric 1024 pref medium
fd00:2005:5:c000::/50 dev eth1 proto kernel metric 256 pref medium
fe80::/64 dev eth0 proto kernel metric 256 pref medium
fe80::/64 dev eth1 proto kernel metric 256 pref medium
default via fe80::a00:27ff:fe30:3c2f dev eth0 proto ra metric 1024 expires 1440sec pref medium
user@host-astra:~$ _
```

Пинги для проверки:

Астра -> Микротик:

Микротик -> Микротик 1:

Микротик 1 -> Базальт-ОС

Базальт -> Астра